END TERM EXAMINATION

FIRST SEMESTER (B.TECH) JANURARY-2024

Paper Code: BS-111 **Subject: Applied Mathematics-I**

Time: 3 Hours **Maximum Marks: 60**

Note: Attempt five questions in all including question no.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1. a) Find $\frac{dy}{dx}$ if $(Cos\ x)^y - (Sin\ y)^x = 0$ (3)

b) Prove that $A^3 - 4A^2 - 3A + 11I = 0$, where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$ (3)

c) Show that the differential $df = x^2dy - (y^2 + xy)dx$ is not exact, but that $dg = (xy^2)^{-1}\,df$ is exact (3)

d) Find the directional derivative of $f(x, y, z) = xy^3 + yz^2$ at the point (2, -1, 1) in the direction of vector $\hat{i} + 2\hat{j} + 3\hat{k}$ (3)

UNIT-I

Q2. a)If $u = f(r)$ where $r^2 = x^2 + y^2 + z^2$, show that $u_{xx} + u_{yy} + u_{zz} = f''(r) + \frac{2}{r}f'(r)$ (4)

b)If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, show that $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$ (4)

c)Locate the stationary points of the function (4)
$f(x, y) = (x^2 - 2y^2)\exp\left[-(x^2 + y^2)/a^2\right]$ where a is a non-zero constant

Q3. a)If $u = x^2 - y^2, v = 2xy$ and $x = r\cos\theta, y = r\sin\theta$, find $\frac{\partial(u, v)}{\partial(r, \theta)}$ (4)

b)The temperature of a point (x,y) on a unit circle is given by $T(x, y) = 1 + xy$. Find the temperature of the two hottest points on the circle. (5)

c)Evaluate $\int_0^\infty \frac{e^{-ax}Sin\ bx}{x}dx$ by applying differentiation under integral sign (3)

UNIT-II

Q4. a)Solve $(1 - x^2)\frac{dy}{dx} + xy = xy^2$ (5)

b)Find the orthogonal trajectories of the family of parabola $y = ax^2$ (2)

c)Solve $\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$ (5)

Q5. a)Find the complete solution for the differential equation $(D^2 - 2D + 1)y = xe^xSinx$ (5)

b)Show that the solution of $y''' - 2y'' + 2y' = 0$ are linearly independent (2)

c)Apply the method of variation of parameters, to solve $x^2y'' + xy' - y = x^2e^x$ (5)

UNIT-III

Q6. a) Determine the rank of the following matrix (4)
$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

b) Discuss the consistency of the following of equations for various values of λ: (5)
$2x_1 - 3x_2 + 6x_3 - 5x_4 = 3$
$x_2 - 4x_3 + x_4 = 1$
$4x_1 - 5x_2 + 8x_3 - 9x_4 = \lambda$

c) Solve $x + 2y + 3z = 20, 7x + 3y + z = 13, x + 6y + 2z = 0$ by Gauss Elimination Method (3)

Q7. a) State Cayley-Hamilton Theorem and Use it to find the ihverse of the matrix (4)
$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$

b)Reduce the quadratic form $4xy + 4yz + 4zx$ in to the canonical form and discuss its nature (3)

c) Solve $x + 2y + 3z = 20$, $7x + 3y + z = 13$, $x + 6y + 2z = 0$ by Cramer's rule (5)

UNIT-IV

Q8. a)Apply Stoke's theorem to evaluate $\int_C (ydx + zdy + xdz)$, where C is the curve of intersection of $x^2 + y^2 + z^2 = b^2$ and $x + z = b$ (8)

b)Show that vector field $\vec{F} = \Delta(x^3 + y^3 + z^3 - 3xyz)$ is irrotational (4)

Q9. a) Find the divergence and curl of $F = xyz\,\hat{i} + 3x^2y\,\hat{j} + (xz^2 - y^2z)\,\hat{k}$ at the point (2,-1,1) (6)

b)Find the curvature and torsion for the curve $x = a\cos t,\ y = a\sin t,\ z = at\cot\alpha$ (6)

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(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] MARCH 2023

Paper Code: BS-111 ***Subject: Applied Mathematics-I***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt **all** questions:-

(a) If $\int_0^1 x^m dx = \frac{1}{m+1}$, thenfind the value of $\int_0^1 x^m (\log x)\, dx$. **(2.5)**

(b) If $\vec{F} = (x+y+1)\hat{i} + \hat{j} - (x+y)\hat{k}$, thencompute the value of $\vec{F}.\,\mathrm{Curl}(\vec{F})$. **(2.5)**

(c) Find the particular integral for the linear differential equation: **(2.5)**

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = \sin 3x.$$

(d) Determine the rank of the matrix $A = \begin{bmatrix} 1 & 3 & 2 & 1 \\ 1 & 2 & 5 & 2 \\ 2 & 1 & 1 & 3 \end{bmatrix}$. **(2.5)**

(e) Applying Gauss divergence theorem, find the value of $\int\int_S \vec{F}.\hat{n}\, ds$, for $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - xz)\hat{j} + (z^2 - yx)\hat{k}$ and S is the cube $0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le z \le 1$. **(2.5)**

(f) Find the stationary values of the function $f(x,y) = x^3y^2(1-x-y)$. **(2.5)**

UNIT-I

Q2 (a) If $u = f(r)$ and $x = r\cos(\theta),\ y = r\sin(\theta)$, then prove that **(7)**

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$$

(b) Evaluate $\int_0^a \frac{\log(1+ax)}{1+x^2}dx$ and hence show that $\int_0^1 \frac{\log(1+x)}{1+x^2}dx = \frac{\pi}{8}\log 2$. **(8)**

Q3 (a) Find the shortest and longest distances from the point $(1,2,-1)$ to the sphere $x^2 + y^2 + z^2 = 24$. **(6)**

(b) If $u = x^2 - y^2,\ v = 2xy$ and $x = r\cos(\theta),\ y = r\sin(\theta)$, then find the JaccobianJ $= \frac{\partial(u,v)}{\partial(r,\theta)}$. **(4)**

(c) If $u = f(e^{y-z}, e^{z-x}, e^{x-y})$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. **(5)**

UNIT-II

Q4 (a)Solve the ordinary differential equation: $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = \log x \sin(\log x)$. **(8)**

(b) Prove that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}}\sin x$, where $J_n(x)$ represents the Bessel function of first kind. **(7)**

Q5 (a) Let the electric equipotential lines (curves of constant potentials) between two concentric cylinders be given by $x^2 + y^2 = c$, where c is the constant. Find their orthogonal trajectories (known as curves of electric force). **(4)**

(b) Solve the ODE: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x \log x$, by variation of parameters. **(6)**

(c) Solve the ODE: $2y\,dx + x(2\log x - y)dy = 0$, by choosing suitable method. **(5)**

P.T.O.